A Additional response experiment

A.1 State-matched response to advocacy

We isolate the one-round behavioral response of the population to controller advocacy while holding the epistemic population fixed. The experiment uses $N = 24$ agents, $q = 1$ social interaction, reasoning depth $L = 2$, support redundancy $r = 6$, recommendation-only control, and an incorrect controller target Z. We fix the sensing and actuation budgets to $q_c = b = 18$ and explicitly initialize three target-support levels,

$$x_0 \in \{0.25, 0.375, 0.50\}, \qquad n_Z(0) \in \{6, 9, 12\}.$$

Within each task, agent identities and initial knowledge sets $K_i(0)$ are identical across all three conditions, giving $\kappa_0 = 0.25$ and $\phi_0 = 0$. Target supporters are selected deterministically and balanced across the populated knowledge strata, with nested sets $Z_6 \subset Z_9 \subset Z_{12}$; all remaining agents are initialized at the correct relation. Thus only the initial social support for Z changes across conditions.

For each x_0, we compare two deterministic one-round interventions: Advocate, in which the controller advocates for Z at exactly $b = 18$ controlled update positions, and NoOp, in which it does not intervene. The two arms use matched episode seeds. We evaluate two tasks with ten matched repetitions per task, yielding 120 one-round episodes. The sensor remains active for logging, but it does not determine the forced action.

The primary quantity is the signed intervention response

$$\chi(x_0) = \mathbb{E}[x_1 - x_0 \mid \mathrm{ADV}, x_0] - \mathbb{E}[x_1 - x_0 \mid \mathrm{NOOP}, x_0], \tag{1}$$

with 95% confidence intervals from 1000 bootstrap resamples of matched episode pairs. Since the action is deterministic within each arm, this experiment estimates a response contrast rather than an action-to-population mutual information.

The measured responses are $\chi(0.25) = 0.163$ (95% CI $[0.119, 0.202]$), $\chi(0.375) = 0.100$ ($[0.060, 0.144]$), and $\chi(0.50) = 0.104$ ($[0.060, 0.148]$). The controller therefore produces a positive causal shift toward its target at all three matched states. However, the largest response occurs at the lowest tested target fraction, and increasing the initial target support from $x_0 = 0.375$ to 0.50 produces no additional gain. The exact $q = 1$ reference predicts the same qualitative ordering—largest leverage at low target support—but substantially larger responses (0.401, 0.334, and 0.268, respectively). Thus, in this controlled one-round setting, LLM advocacy is effective but markedly weaker than the idealized copying reference, and there is no evidence that additional pre-existing social support amplifies the intervention response.

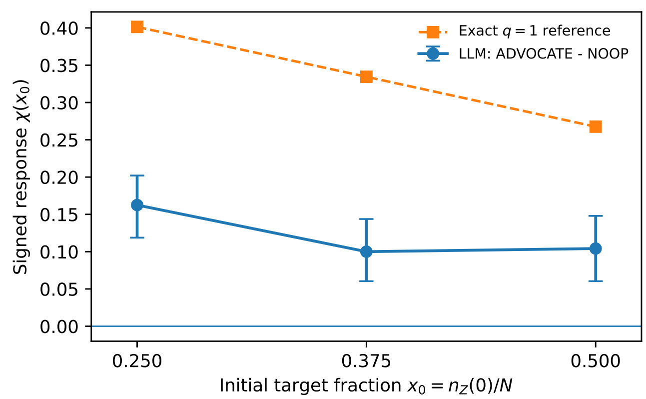

Figure 1: One-round state-matched response to advocacy. Points show the empirical Advocate-minus-NoOp response with 95% matched-pair bootstrap intervals. The dashed curve is the exact $q = 1$ reference at $N = 24$ and $b = 18$, $\chi_{q=1}(x_0) = (1 - x_0)[1 - (1 - 1/N)^b]$. Advocacy produces a positive response at every imposed initial target fraction, while the response does not increase with pre-existing target support over the tested range.